

Your INPUT goes here!

```

17 key[0] = 0;
18 parent[0] = -1;
19
20 for (int count = 0; count < V - 1; count++) {
21     int min = INF, u;
22
23     for (int v = 0; v < V; v++)
24         if (mstSet[v] == false && key[v] < min)
25             min = key[v], u = v;
26
27     mstSet[u] = true;
28
29     for (int v = 0; v < V; v++)
30         if (graph[u][v] && mstSet[v] == false && graph[u][v] < key[v])
31             parent[v] = u, key[v] = graph[u][v];
32 }
33
34 for (int i = 1; i < V; i++)
35     sum += graph[i][parent[i]];
36
37 printf("Total Weight: %d\n", sum);

```

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QUESTIONS

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MST using Prim's A

MST USING PRIM'S ALGORITHM

Construct the MST using Prim's Algorithm and display its total weight.

Input Matrix:

0 4 3 0
4 0 2 5
3 2 0 7
0 5 7 0

PROGRAM EDITOR

C

Run

Save

```
30         if (graph[u][v] && mstSet[v] == false && graph[u][v] < key[v])
31             parent[v] = u, key[v] = graph[u][v];
32     }
33
34     for (int i = 1; i < V; i++)
35         sum += graph[i][parent[i]];
36
37     printf("Total Weight: %d\n", sum);
38 }
39
40 int main() {
41     int graph[V][V] = {
42         {0, 4, 3, 0},
43         {4, 0, 2, 5},
44         {3, 2, 0, 7},
45         {0, 5, 7, 0}
46     };
47
48     primMST(graph);
49     return 0;
50 }
```

INPUT

Your INPUT goes here!